

A better blue pixel

Can TADF deliver efficiency and lifetime?

Follow an excitation.
Then judge the device.

Efficiency

Colour

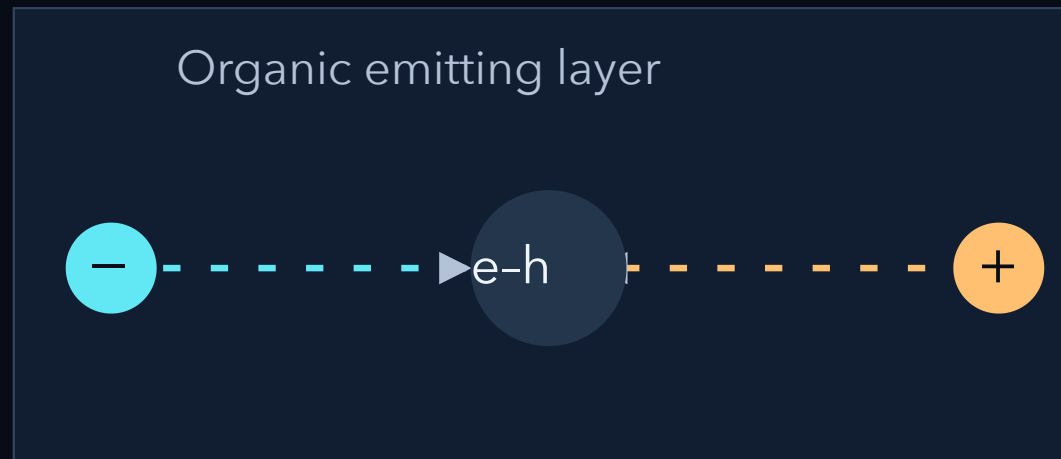
Operating life



One generic subpixel. Three requirements.

One excitation. Four spin states.

Electrical recombination creates singlets and triplets.



Charge → exciton → light

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

1

singlet

$$P(S) = \frac{1}{4}$$

3

triplets

$$P(T) = \frac{3}{4}$$

Triplet states: $m = -1, 0, +1$.

Spin statistics assume uncorrelated formation.

Spin controls which radiative routes are allowed.

Triplets already have competitors

TADF competes with other ways of using triplet excitations.

Triplet fusion



Two triplets can make one emissive singlet.

Ideal ceiling: 62.5%

Phosphorescence



Spin-orbit mixing opens triplet-character emission.

Ideal ceiling: 100%

TADF



Recover triplets, then emit fluorescence.

Ideal ceiling: 100%

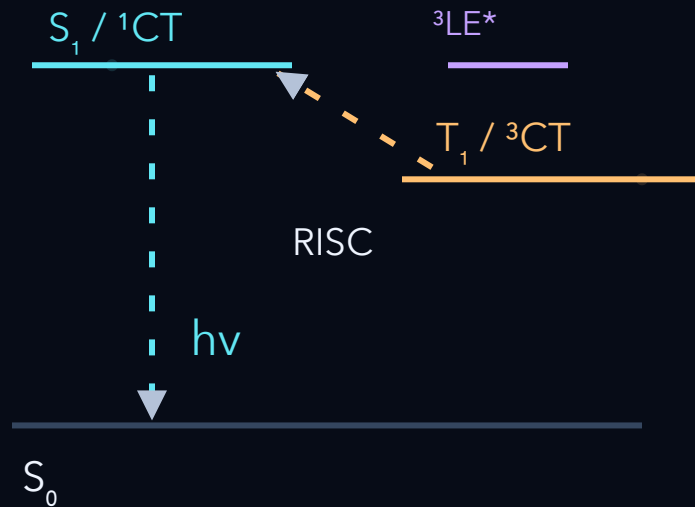
Singlet-only reference: 25%.

Ideal exciton-use limits per emitting unit, with successful conversion and no competing loss.

These are not measured device efficiencies. Electrically tandem devices need separate accounting.

Recovery needs a route—and a rate

RISC: reverse intersystem crossing



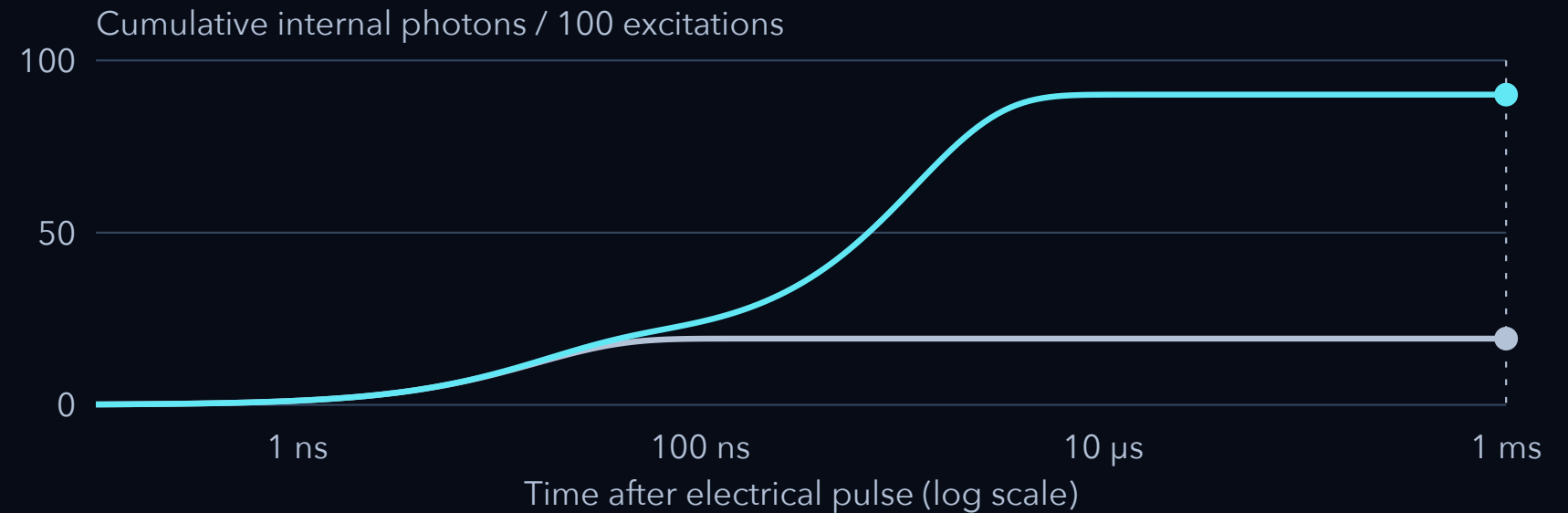
*Spin-vibronic assistance: an example, not universal.

ΔE_{ST} : energetic access

$k_B T \approx 0.026$ eV at 300 K

$$k_{RISC} \propto |\langle S_1 | \hat{H}_{eff} | T_1 \rangle|^2 F_{vib}(T)$$

Schematic: coupling $\times$ energetic/vibrational factors



– Singlet-only reference – RISC on

Reference **19.2** photons RISC on **90.1** photons

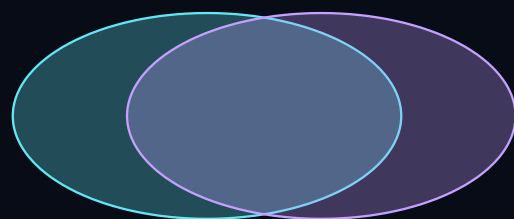
100 excitations; deterministic ensemble. Logarithmic physical time, slowed playback.
Internal photons shown; identical illustrative rates except k_{RISC} .

A small gap is only half the design

Make recovery accessible without making emission weak.

$$\Delta E_{ST} \approx 2K_{HL}$$

Exchange splitting: simple common-orbital picture

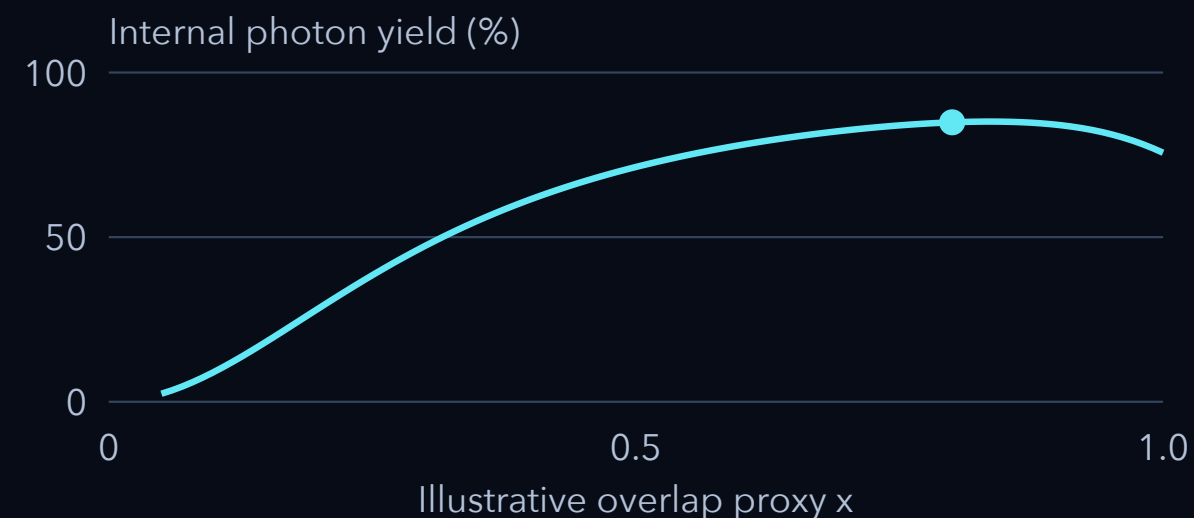


Orbital-overlap illustration • not a wavefunction calculation

$$k_r \propto \omega^3 |\mu_{S_0 S_1}|^2$$

$$\mu_{S_0 S_1} = \langle S_0 | \hat{\mu} | S_1 \rangle$$

Exchange and optical transition strength
are different quantities.



Yield 84.9% · E_a 0.135 eV · k_{RISC} 5.35e+5 s⁻¹

Toy design map; not a torsion angle, fitted molecule or universal optimum.
Fixed- E_a presets change only effective coupling.

Let different molecules do different jobs

Hyperfluorescence: harvest, transfer, emit.



Harvesting and colour generation
can be assigned separately.

Energy transfer competes with loss.
Unwanted triplet transfer remains a design risk.
Animation is a pathway illustration, not a rate simulation.

Count the photons that leave

A good emitter is one factor in a good device.

$$\text{EQE} \approx \gamma \cdot \Phi_{\text{EL}} \cdot \eta_{\text{out}}$$

Form excitons

Charge balance and
exciton formation

Make photons

Effective internal yield:
all branching and recycling

Extract light

Photons that escape
the optical stack

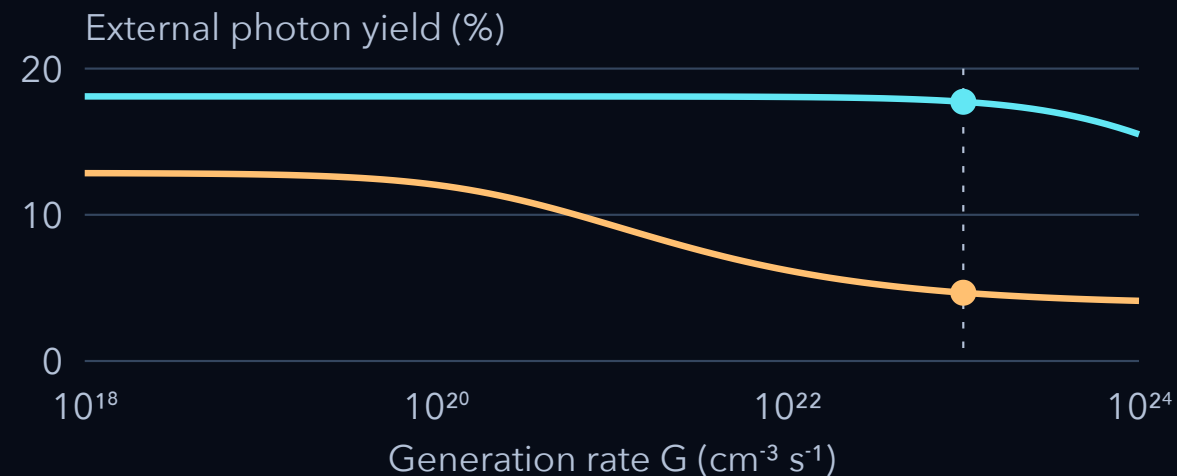


Power also depends on voltage and spectrum.

High-density loss is not operating life

Compare a kinetic model with an experimental efficiency measurement.

Illustrative steady-state model



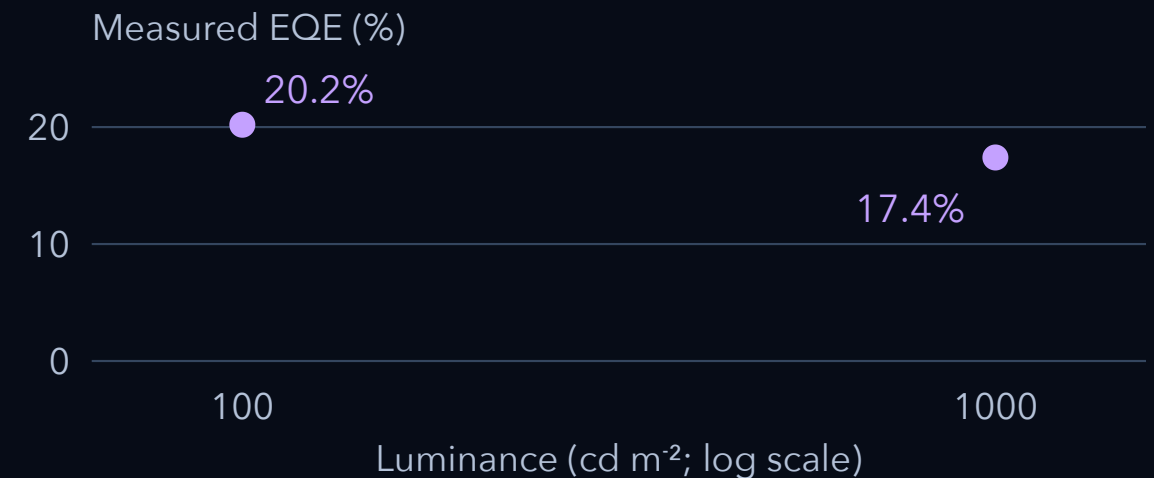
Slower RISC 4.7% · Faster RISC 17.7%

kRISC: 2×10^4 vs $2 \times 10^6 \text{ s}^{-1}$ · $G = 10^{23.0}$

βT^2 is net triplet quenching. No degradation model.

Generation rate is not calibrated to luminance.

Reported TMCz-BO device data



Two reported points • Table 2, Kim et al. (2020)

CIE (0.14, 0.18); peak EQE 20.7%.

Re-plotted values; no interpolation claimed.

Operating life needs a separate
luminance-decay test.

Evidence has a colour and a stage

Device demonstration, panel verification and shipped products answer different questions.

Idemitsu BLUE DEVICE • 2022	Triplet-fusion-assisted fluorescence 14% EQE at 10 mA cm ⁻² ; CIE (0.14, 0.08)
LG Display PANEL VERIFICATION • 2025	Fluorescent/phosphorescent hybrid blue Company reports ≈15% lower power vs its reference
Kyulux / WiseChip PMOLED PRODUCT • 2023	5.5-inch Hyperfluorescence display Real product in this class; not blue AMOLED qualification

2026 Kyulux / SK Materials JNC: green-material licence;
2026 samples and 2027 production are stated targets.



Product photo: WiseChip, via Kyulux (2023).

Match the claim to its
colour, architecture and stage.

Value depends on the whole panel life

Lower power, durable use and saleable yield must survive together.

Environmental value

Lower panel power



Less use-phase energy

Longer useful life



Fewer replacements, if realised

Manufacturing value

Reliable process + good yield



Lower cost per saleable panel

Material price alone is insufficient.
Include qualification, process consistency
and the cost of rejected panels.

No automatic carbon saving follows from “metal-free”. The comparison needs a defined device and use case.

Adopt the device, not the headline

TADF offers a route to useful singlets. Adoption needs three tests.

- 1 Lower electrical power at matched colour and output
 - 2 Measured operating life that meets the customer's target
 - 3 Repeatable manufacture at acceptable cost
-

Better physics becomes a better product when all three hold.

Model equations and conservation

Q&A • the model behind the moving picture

$$\dot{S} = G/4 + rT - K_S S$$

$$\dot{T} = 3G/4 + k_{ISC} S - (r + k_{nr,T}) T$$

$$\dot{F} = k_r S$$

$$\dot{Q} = k_{nr,S} S + k_{nr,T} T$$

$$K_S = k_r + k_{nr,S} + k_{ISC}$$

Pulse: $G=0$, $S(0)=25$, $T(0)=75$.

$S+T+F+Q=100$; exact linear-system solution.

$$\Phi_{EL} = a(1/4 + 3/4 q) / (1 - bq)$$

$$a = k_r / K_S, \quad b = k_{ISC} / K_S, \quad q = r / (r + k_{nr,T})$$

Default illustrative rates / s^{-1}

$$k_r = 5 \times 10^7$$

$$k_{nr,S} = 5 \times 10^6$$

$$k_{ISC} = 1 \times 10^7$$

$$k_{nr,T} = 1 \times 10^4$$

$$r = 0 \text{ versus } 1 \times 10^6$$

Final internal yield: 19.23% versus 90.07%.

$\eta_{out}=0.20$ would give 3.85% versus 18.01%.

These are model values, not commercial gains.

What the sliders assume

Q&A • inspect the assumptions before interpreting an optimum

Illustrative molecular design

$$r = A_0 c^2 \exp(-E_a/k_B T)$$

$A_0 = 10^8 \text{ s}^{-1}$; temperature = 300 K.

Fixed gap: $E_a = 0.10 \text{ eV}$; $c = 0.1$ or 1 .

$$E_a(x) = 0.02 + 0.18x^2 \text{ eV}$$

$$k_r(x) = 5 \times 10^7 x^2 \text{ s}^{-1}$$

x is an overlap proxy, not a molecular angle.

The toy diagram sets $E_a = \Delta E_{ST}$; this is not universal.

No detailed-balance or quantum-chemistry fit.

Density and net quenching

$$\dot{T} \rightarrow \dot{T} - \beta T^2$$

$$T_{ss} = 2B/(D + \sqrt{D^2 + 4\beta B})$$

$$B = (\frac{3}{4} + k_{ISC}/(4K_S))G$$

$$D = r + k_{nr,T} - k_{ISC}r/K_S$$

$$S_{ss} = (G/4 + rT_{ss})/K_S$$

$$\text{External yield} = 0.2 k_r S_{ss}/G$$

$$\beta = 10^{-12} \text{ cm}^3 \text{ s}^{-1}; G = 10^{18} - 10^{24} \text{ cm}^{-3} \text{ s}^{-1}.$$

$$\text{Compare } r = 2 \times 10^4 \text{ and } 2 \times 10^6 \text{ s}^{-1}.$$

β represents net depletion, not every TTA event.

Lifetime, energy and cost

Q&A • keep the endpoint and the denominator visible

$$E_{\gamma} = hc/\lambda$$

450 nm → 2.76 eV

630 nm → 1.97 eV

$$L(LT_x)/L(0) = x/100$$

LT95 is time to 95% of starting luminance.

State initial luminance, drive mode and environment.

Idemitsu reports LT95 >400 h at 50 mA cm⁻²;

its 14% EQE value uses 10 mA cm⁻².

$$\text{Cost per good panel} \approx C_{\text{attempt}}/Y$$

Illustrative cost units:

$$100/0.90 = 111.1$$

$$95/0.80 = 118.8$$

Lower attempt cost can still mean higher unit cost.

Dated context only: Apple's iPhone 16 Pro 128 GB report gives 66 kg CO₂e, with 81% production and 17% product use (rounded, whole phone). This is not a TADF or display-specific carbon estimate.

Sources and image credit

Q&A • primary papers and bounded company evidence

[Etherington et al. \(2016\) · spin-vibronic coupling](#)

[Stavrou et al. \(2024\) · hyperfluorescence](#)

[LG Display \(2025\) · hybrid blue panel](#)

[Kyulux \(2026\) · green material licence](#)

[Kim et al. \(2020\) · TMCz-BO and roll-off](#)

[Idemitsu \(2022\) · blue TTF device](#)

[Kyulux / WiseChip \(2023\) · PMOLED product + photo](#)

[Apple \(Sept 2024\) · whole-phone lifecycle example](#)

The product photo is credited to WiseChip via Kyulux. All other visuals are explanatory diagrams or original model plots. Reported experimental points are re-plotted from Kim et al., Table 2; no digitised full curve is implied.